

INSERTION SORT :

Insertion Sort is a simple sorting algorithm that sorts an array by taking one element at a time and inserting it into its correct position in the already sorted part of the array.

ALGORITHM:

1. Start from the second element of the array.
2. Store the current element as the key.
3. Compare the key with the previous elements.
4. Shift all larger elements one position to the right.
5. Insert the key into its correct position.
6. Repeat the process until all elements are sorted.

Time Complexity:

Case	Time Complexity
Best Case (Already Sorted)	$O(n)$
Average Case	$O(n^2)$
Worst Case (Reverse Sorted)	$O(n^2)$

Space Complexity:

Space Complexity = $O(1)$

OUTPUT:

```
Enter number of elements: 6
Enter the elements:
5
7
2
1
4
9
Sorted elements are:
1 2 4 5 7 9

...Program finished with exit code 0
Press ENTER to exit console.
```